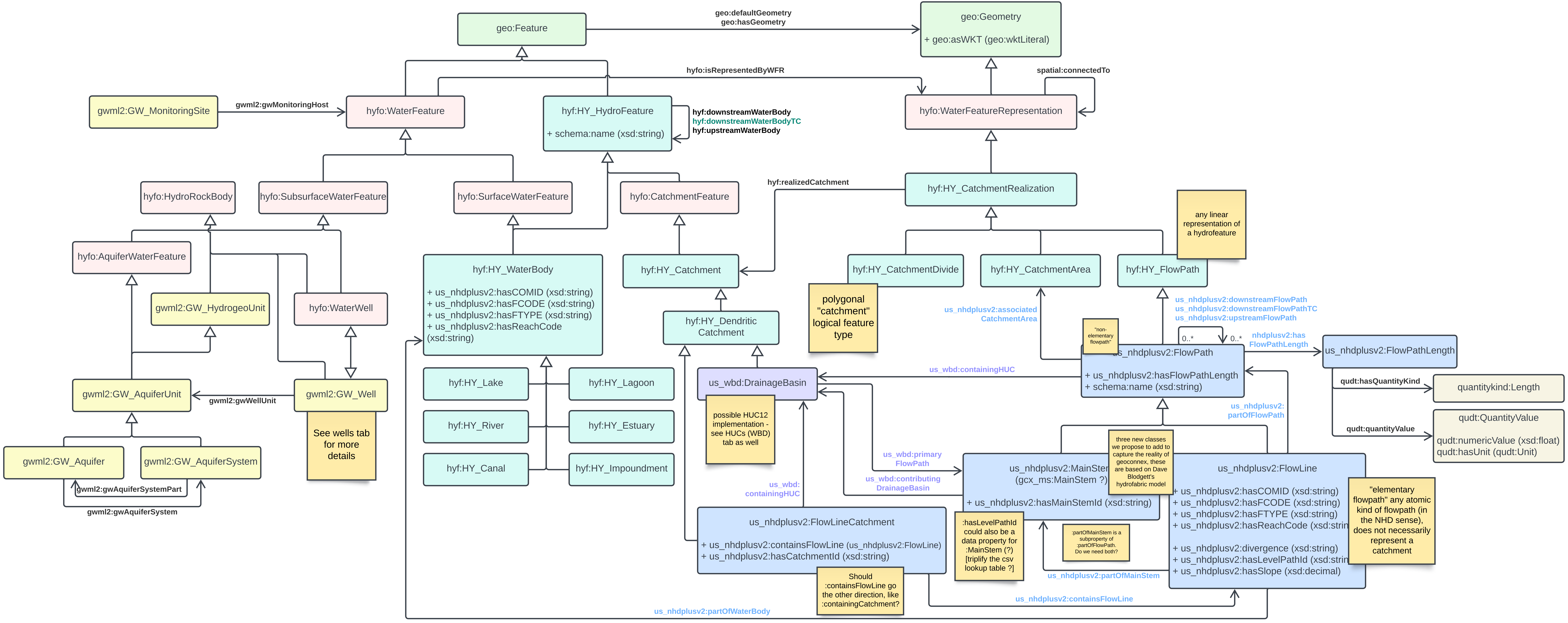
geo:Feature
geo:defaultGeometry
geo:hasGeometry
geo:Geometry
+ geo:asWKT (geo:wktLiteral)
hyfo:isRepresentedByWFR
spatial:connectedTo
gwml2:GW_MonitoringSite
gwml2:gwMonitoringHost
hyfo:WaterFeature
hyf:HY_HydroFeature
+ schema:name (xsd:string)
hyf:downstreamWaterBody
hyf:downstreamWaterBodyTC
hyf:upstreamWaterBody
hyfo:WaterFeatureRepresentation
hyf:realizedCatchment
hyf:HY_CatchmentRealization
hyfo:HydroRockBody
hyfo:SubsurfaceWaterFeature
hyfo:SurfaceWaterFeature
hyfo:CatchmentFeature
hyfo:AquiferWaterFeature
hyf:HY_WaterBody
+ us_nhdplusv2:hasCOMID (xsd:string)
+ us_nhdplusv2:hasFCODE (xsd:string)
+ us_nhdplusv2:hasFTYPE (xsd:string)
+ us_nhdplusv2:hasReachCode (xsd:string)
hyf:HY_Catchment
hyf:HY_CatchmentDivide
hyf:HY_CatchmentArea
hyf:HY_FlowPath
gwml2:GW_HydrogeoUnit
hyfo:WaterWell
hyf:HY_Dendritic Catchment
us_nhdplusv2:associated CatchmentArea
us_nhdplusv2:downstreamFlowPath
us_nhdplusv2:downstreamFlowPathTC
us_nhdplusv2:upstreamFlowPath
0..*
0..*
nhdplusv2:has FlowPathLength
us_nhdplusv2:FlowPath
+ us_nhdplusv2:hasFlowPathLength
+ schema:name (xsd:string)
us_nhdplusv2:FlowPathLength
qudt:hasQuantityKind
quantitykind:Length
qudt:quantityValue
qudt:QuantityValue
qudt:numericValue (xsd:float)
qudt:hasUnit (qudt:Unit)
us_wbd:containingHUC
us_wbd:DrainageBasin
gwml2:GW_AquiferUnit
gwml2:gwWellUnit
gwml2:GW_Well
hyf:HY_Lake
hyf:HY_Lagoon
hyf:HY_River
hyf:HY_Estuary
hyf:HY_Canal
hyf:HY_Impoundment
us_nhdplusv2: partOfFlowPath
us_wbd:primary FlowPath
us_wbd:contributing DrainageBasin
us_nhdplusv2:MainStem (gcx_ms:MainStem ?)
+ us_nhdplusv2:hasMainStemId (xsd:string)
us_nhdplusv2:FlowLine
+ us_nhdplusv2:hasCOMID (xsd:string)
+ us_nhdplusv2:hasFCODE (xsd:string)
+ us_nhdplusv2:hasFTYPE (xsd:string)
+ us_nhdplusv2:hasReachCode (xsd:strin
+ us_nhdplusv2:divergence (xsd:string)
+ us_nhdplusv2:hasLevelPathId (xsd:strin
+ us_nhdplusv2:hasSlope (xsd:decimal)
gwml2:GW_Aquifer
gwml2:GW_AquiferSystem
gwml2:gwAquiferSystemPart
gwml2:gwAquiferSystem
us_wbd: containingHUC
us_nhdplusv2:FlowLineCatchment
+ us_nhdplusv2:containsFlowLine (us_nhdplusv2:FlowLine)
+ us_nhdplusv2:hasCatchmentId (xsd:string)
us_nhdplusv2:partOfMainStem
us_nhdplusv2:containsFlowLine
us_nhdplusv2:partOfWaterBody

any linear representation of a hydrofeature

polygonal "catchment" logical feature type

"non-elementary flowpath"

possible HUC12 implementation - see HUCs (WBD) tab as well

See wells tab for more details

three new classes we propose to add to capture the reality of geoconnex, these are based on Dave Blodgett's hydrofabric model

"elementary flowpath" any atomic kind of flowpath (in the NHD sense), does not necessarily represent a catchment

:hasLevelPathId could also be a data property for :MainStem (?) [triplify the csv lookup table ?]

:partOfMainStem is a subproperty of :partOfFlowPath. Do we need both?

Should :containsFlowLine go the other direction, like :containingCatchment?